

Grading Selection from Adjoint Spectral Activity in $\mathfrak{sl}(n, \mathbb{R})$

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Abstract

For the Lie algebra $\mathfrak{sl}(n, \mathbb{R})$ equipped with a diagonal Cartan element T having eigenvalue clusters $C_1, \dots, C_m$, we show that the functional $\tilde{S}_g[P] = \text{Tr}(\sigma_P \cdot g(\text{ad}_T))$ over symmetric involutions P , for any even $g : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ with $g(0) = 0$ and $g(\lambda) > 0$, is minimised at a cluster-coherent partition: each eigenvalue cluster is assigned entirely to one side. The proof shows that $g(0) = 0$ eliminates intra-cluster contributions, making the functional bilinear in cluster magnetisations, so splitting a cluster never helps — provided the interaction weights factor through eigenvalue clusters. This factorisation holds for all four classical Lie algebra families (Corollary 8); for exceptional algebras (G_2 , F_4 , E_{6-8}), eigenvalue clusters can contain roots of multiple lengths and the optimal grading may split clusters (Proposition 9). The optimisation reduces to a weighted max-cut on the complete graph K_m with edge weights $w_{ij} = g(|t_i - t_j|) n_i n_j$. For binary splits ($m = 2$), the minimiser is unique and places the minority cluster in the P -negative eigenspace; for $T_{B-L} = \text{diag}(1/3, 1/3, 1/3, -1)$, the selected grading has $(3, 1)$ shape. If $f(0) > 0$, the zero-mode contribution can change the selected grading (Proposition 7).

The result is verified across $\mathfrak{sl}(3, \mathbb{R})$ – $\mathfrak{sl}(6, \mathbb{R})$ (nine Cartan structures including binary and ternary splits), six spectral functionals, $O(n)$ frame invariance, perturbation stability, and all four classical Lie algebra types.

We emphasise that the theorem selects a grading of the *internal* carrier space of $\mathfrak{sl}(4, \mathbb{R})$, not of physical spacetime. In light of the Coleman–Mandula theorem, identifying this internal $(3, 1)$ decomposition with Lorentzian spacetime signature would require an additional bridging mechanism not provided here.

1 Setup

Let $\mathfrak{g} = \mathfrak{sl}(4, \mathbb{R})$ with basis $\{E_\alpha\}_{\alpha=1}^{15}$ and trace inner product $\langle X, Y \rangle = \text{Tr}(XY)$.

Definition 1 (Distinguished generator). A Cartan element $T \in \mathfrak{g}$ is a traceless diagonal matrix $T = \text{diag}(t_1, \dots, t_4)$ with $\sum_i t_i = 0$. Its adjoint ad_T acts on $\mathfrak{g}$ by $\text{ad}_T(X) = [T, X]$, with eigenvalues $\{t_i - t_j : i \neq j\} \cup \{0\}$.

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Definition 2 (Symmetric involution and grading). A symmetric involution on $\mathbb{R}^4$ is a matrix $P \in \text{GL}(4, \mathbb{R})$ with $P^2 = I$ and $P = P^T$. It induces a grading on $\mathfrak{g}$ via $\sigma_P(X) = PXP^{-1}$:

$$\mathfrak{g} = \mathfrak{g}_+ \oplus \mathfrak{g}_-, \quad \mathfrak{g}_\pm = \{X \in \mathfrak{g} : PXP^{-1} = \pm X\}.$$

The space of symmetric involutions with fixed signature (p, q) (p positive eigenvalues, $q = 4-p$ negative) is the Grassmannian $\text{Gr}(q, 4)$.

The physically relevant signatures are: $(4, 0)$ (Euclidean / trivial), $(3, 1)$ or $(1, 3)$ (Lorentzian), and $(2, 2)$ (split / ultrahyperbolic).

Definition 3 (Graded spectral functional). For an even function $g : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ with $g(0) = 0$ and $g(\lambda) > 0$ for $\lambda \neq 0$, define

$$\tilde{\mathcal{S}}_g[P] = \text{Tr}(\sigma_P \cdot g(\text{ad}_T)), \quad (1)$$

where σ_P is the adjoint representation of the involution $X \mapsto PXP^{-1}$ (an $(n^2-1) \times (n^2-1)$ matrix with eigenvalues ± 1), and $g(\text{ad}_T)$ is defined via the spectral theorem applied to ad_T . The condition $g(0) = 0$ ensures that the functional depends only on the active (nonzero-eigenvalue) sector of ad_T , eliminating the zero-mode contribution.

The functional $\tilde{\mathcal{S}}_g$ is invariant under simultaneous $\text{O}(n)$ conjugation: $T \rightarrow GTG^T$, $P \rightarrow GPG^T$.

2 Spectral structure of $\text{ad}_{T_{B-L}}$

For $T_{B-L} = \text{diag}(1/3, 1/3, 1/3, -1)$:

Proposition 4 (Spectral decomposition of $\text{ad}_{T_{B-L}}$). *The adjoint operator $\text{ad}_{T_{B-L}}$ on $\mathfrak{sl}(4, \mathbb{R})$ has spectrum $\{0, +4/3, -4/3\}$ with multiplicities $(9, 3, 3)$.*

- The 9-dimensional kernel consists of the Cartan generators $\{H_k\}$ and the purely spatial generators $\{A_{ij}, S_{ij}\}$ with $i, j \in \{0, 1, 2\}$ (quark-quark interactions).
- The 6-dimensional active sector consists of the quark-lepton mixing generators $\{A_{i3}, S_{i3}\}$ for $i \in \{0, 1, 2\}$, split equally between the $+4/3$ and $-4/3$ eigenspaces.

Since $\text{ad}_{T_{B-L}}$ has only two distinct eigenvalue magnitudes (0 and $4/3$), any even function f acts as:

$$f(\text{ad}_{T_{B-L}}) = f(0) \cdot \Pi_{\text{ker}} + f(4/3) \cdot \Pi_{\text{active}}, \quad (2)$$

where Π_{ker} and Π_{active} are the orthogonal projectors onto the 9- and 6-dimensional subspaces.

3 The selection theorem

We first establish the combinatorial structure of admissible parity assignments, then prove the selection theorem.

Lemma 5 (Parity induced by index partitions). *Since $\tilde{\mathcal{S}}_g$ is invariant under simultaneous $\text{O}(n)$ conjugation of (T, P) , we may fix a basis in which T is diagonal and restrict to symmetric involutions P that are diagonal in this basis without loss of generality.*

In such a basis, $P = \text{diag}(\epsilon_1, \dots, \epsilon_n)$ with $\epsilon_i = \pm 1$, corresponding to a partition

$$\{1, \dots, n\} = I_+ \sqcup I_-, \quad I_\pm = \{i : \epsilon_i = \pm 1\}.$$

Under the induced adjoint action $\sigma_P(X) = PXP^{-1}$, the standard generators E_{ab} of $\mathfrak{sl}(n, \mathbb{R})$ (with matrix entries $(E_{ab})_{cd} = \delta_{ac}\delta_{bd}$ for the off-diagonal generators, and $(H_k)_{ii} = \delta_{ik} - \delta_{i,k+1}$ for the Cartan) satisfy

$$P E_{ab} P^{-1} = \epsilon_a \epsilon_b E_{ab}. \quad (3)$$

Hence the parity eigenvalue of E_{ab} under σ_P is

$$p_{ab} = \begin{cases} +1 & \text{if } a, b \in I_+ \text{ or } a, b \in I_-, \\ -1 & \text{if } a \in I_+, b \in I_- \text{ or } a \in I_-, b \in I_+. \end{cases}$$

The map $\text{Part}(\{1, \dots, n\}) \rightarrow \{-1, +1\}^{n^2-1}$ sending a partition to its induced parity assignment is injective modulo exchange $I_+ \leftrightarrow I_-$ (which corresponds to $P \leftrightarrow -P$ and produces identical conjugation $\sigma_P = \sigma_{-P}$).

Proof. Every symmetric involution is diagonalisable with eigenvalues ± 1 , hence conjugate to a diagonal sign matrix by an orthogonal transformation. The parity rule (3) follows from $(PXP^{-1})_{cd} = \epsilon_c \epsilon_d X_{cd}$. For injectivity: if two partitions (I_+, I_-) and (J_+, J_-) induce the same parity on all generators, then $\epsilon_a \epsilon_b = \epsilon'_a \epsilon'_b$ for all a, b , which implies $\epsilon_a = \epsilon'_a$ for all a (up to a global sign $\epsilon_a \rightarrow -\epsilon_a$, i.e. exchange of $I_+ \leftrightarrow I_-$). $\square$

Theorem 6 (Grading selection from adjoint activity). *Let $\mathfrak{g}$ be a real Lie algebra with Cartan element T whose adjoint eigenvalues define clusters $C_1, \dots, C_m$. Let $g : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ be even with $g(0) = 0$ and $g(\lambda) > 0$ for $\lambda \neq 0$. Suppose the interaction weights $w_{ab} = g(|\lambda_a - \lambda_b|)$ factor through eigenvalue clusters: $w_{ab} = W_{\alpha\beta}$ whenever $a \in C_\alpha$ and $b \in C_\beta$. Consider the functional*

$$\tilde{\mathcal{S}}_g(P) = \text{Tr}(\sigma_P g(\text{ad}_T))$$

over symmetric involutions P . Then:

- (i) (Cluster coherence.) *The minimum is attained at a partition that assigns each eigenvalue cluster entirely to I_+ or entirely to I_- . Splitting a cluster between the two sides never reduces $\tilde{\mathcal{S}}_g$.*
- (ii) (Reduction to weighted max-cut.) *Under cluster-coherent partitions, the functional reduces to*

$$\tilde{\mathcal{S}}_g = \sum_{i < j} 2 w_{ij} n_i n_j \sigma_i \sigma_j, \quad (4)$$

where $w_{ij} = g(|t_i - t_j|) > 0$ are inter-cluster weights and $\sigma_i = \pm 1$ is the sign assigned to cluster C_i . The minimiser is the sign assignment $\{\sigma_i\}$ that maximises the total cut weight $\sum_{(i,j) \text{ cut}} w_{ij} n_i n_j$ on the complete weighted graph K_m .

- (iii) (Binary split.) *If $m = 2$ with $|C_1| = k < n/2$, the unique minimiser places the minority cluster in I_- , giving $\tilde{\mathcal{S}}_g^{\min} = -2k(n-k)g(\lambda)$.*
- (iv) (Structural stability.) *Under perturbation $T \rightarrow T + \varepsilon H$, the minimiser changes only at eigenvalue-crossing transitions where the cluster structure of T changes.*

In particular, for $T_{B-L} = \text{diag}(1/3, 1/3, 1/3, -1)$ ($m = 2$, binary split), the selected grading has $(3, 1)$ shape.

Proposition 7 (Zero-mode contribution). *If $f : \mathbb{R} \rightarrow \mathbb{R}$ is even with $f(0) > 0$ and $f(\lambda) > 0$ for $\lambda \neq 0$, the functional $\mathcal{S}_f[P] = \text{Tr}(\sigma_P f(\text{ad}_T))$ includes a zero-mode contribution $f(0) \text{Tr}(\sigma_P|_{\ker \text{ad}_T})$ that competes with the active-sector term. In this case the selected grading need not coincide with the cluster-coherent rule; the relative magnitudes of $f(0)$ and $f(\lambda)$ determine whether the zero-mode contribution changes the minimiser.*

Proof of Theorem 6. We prove parts (i)–(iii); part (iv) is verified numerically (Section 4).

Step 1 (Magnetisation decomposition). By Lemma 5, any admissible involution P corresponds to a partition $\{1, \dots, n\} = I_+ \sqcup I_-$ with parity $p_{ab} = \epsilon_a \epsilon_b$. For each eigenvalue cluster C_i , define $k_i = |C_i \cap I_-|$ and the magnetisation $m_i = n_i - 2k_i$.

The active generator pairs are those connecting different clusters. Since $g(0) = 0$, intra-cluster pairs contribute nothing. For an inter-cluster pair (a, b) with $a \in C_i, b \in C_j$ ($i \neq j$), the contribution is $p_{ab} g(|t_i - t_j|)$.

Summing over all ordered pairs within the C_i – C_j block: the number of same-sign pairs minus cross-sign pairs is $(n_i - 2k_i)(n_j - 2k_j) = m_i m_j$ (algebraic identity). Hence:

$$\tilde{\mathcal{S}}_g = \sum_{i < j} 2g(|t_i - t_j|) m_i m_j. \quad (5)$$

Step 2 (Cluster coherence — proof of part (i)). The key observation: $g(0) = 0$ eliminates all m_i^2 terms from $\tilde{\mathcal{S}}_g$. The functional (5) is therefore *bilinear* in the magnetisations $(m_1, \dots, m_m)$.

For fixed $\{m_j\}_{j \neq i}$, $\tilde{\mathcal{S}}_g$ is *linear* in m_i :

$$\tilde{\mathcal{S}}_g = m_i \cdot \left[\sum_{j \neq i} 2w_{ij} m_j \right] + (\text{terms independent of } m_i),$$

where $w_{ij} = g(|t_i - t_j|) > 0$. A linear function on the interval $m_i \in [-n_i, n_i]$ is minimised at one of the endpoints: $m_i = +n_i$ (all of C_i in I_+ , i.e. $k_i = 0$) or $m_i = -n_i$ (all of C_i in I_- , i.e. $k_i = n_i$). Interior values ($0 < k_i < n_i$, splitting the cluster) are *never* optimal.

Since this argument applies to each cluster independently, the global minimum is attained at a *corner* of the box $\prod_i [-n_i, n_i]$, i.e. a cluster-coherent partition.

Step 3 (Reduced problem — proof of part (ii)). With cluster-coherent partitions, set $\sigma_i = \pm 1$ according to whether $C_i \subseteq I_+$ or $C_i \subseteq I_-$. Then $m_i = \sigma_i n_i$ and (5) becomes

$$\tilde{\mathcal{S}}_g = \sum_{i < j} 2w_{ij} n_i n_j \sigma_i \sigma_j.$$

Minimising is equivalent to maximising the total cut weight $\sum_{(i,j) \text{ cut}} w_{ij} n_i n_j$ on the complete weighted graph K_m , which is (4).

Step 4 (Binary case — proof of part (iii)). For $m = 2$ with clusters C_1 ($|C_1| = k$) and C_2 ($|C_2| = n - k$), the only nontrivial partition ($\sigma_1 = -\sigma_2$) has cut weight $w_{12} k(n - k)$. The trivial partition ($\sigma_1 = \sigma_2$) cuts nothing. Hence the unique minimiser places C_1 entirely in I_- and C_2 in I_+ (or vice versa), giving

$$\tilde{\mathcal{S}}_g^{\min} = -2k(n - k)g(\lambda).$$

Step 5 (Uniqueness). For $m = 2$: the max-cut requires $\sigma_1 \neq \sigma_2$, which is unique up to global sign. For $m \geq 3$: the max-cut on K_m may have multiple optimal sign assignments if the weight matrix has special symmetries (e.g. equal-weight clusters). In the generic case (all $w_{ij} n_i n_j$ distinct), the max-cut is unique up to global sign.

In all tested cases ($m \leq 3$ across $\mathfrak{sl}(3, \mathbb{R})$ – $\mathfrak{sl}(6, \mathbb{R})$, nine Cartan structures), the optimal sign assignment is unique up to global sign and consistent with the pattern: the cluster(s) most separated from the “centre of mass” of the eigenvalues are assigned to I_- . $\square$

Remark (Uniqueness). The minimiser is unique up to global sign ($I_+ \leftrightarrow I_-$) and permutations within equal-eigenvalue clusters of T . For T_{B-L} with clusters $\{1/3\}^3$ and $\{-1\}^1$, the minority cluster has size 1 (no internal permutations), so the selected grading $P = \text{diag}(1, 1, 1, -1)$ is unique up to $\text{SO}(3)$ rotations of the quark colour indices.

Corollary 8 (Classical types satisfy weight factorisation). *For classical Lie algebras (A_n, B_n, C_n, D_n) in their defining representations, all generators connecting two eigenvalue clusters of ad_T carry the same weight $g(|t_\alpha - t_\beta|)$. The factorisation condition of Theorem 6 is therefore satisfied, and cluster coherence holds.*

Proof. In type A_n ($\mathfrak{sl}(n, \mathbb{R})$), adjoint eigenvalues are differences $t_a - t_b$ of defining-representation eigenvalues. Generators E_{ab} and $E_{a'b'}$ with $a, a' \in C_\alpha$ and $b, b' \in C_\beta$ have the same eigenvalue difference $t_\alpha - t_\beta$, hence the same weight. Types B_n, C_n, D_n have uniform root length within each eigenvalue cluster (long roots connect different clusters, short roots lie within clusters), giving the same factorisation. $\square$

Proposition 9 (Failure of cluster coherence in G_2). *There exist Cartan elements in G_2 whose eigenvalue clusters contain roots of both short and long length. For such elements, the interaction weights do not factor through eigenvalue clusters, and the optimal grading splits clusters.*

Specifically, for $H = (2, -1, -1)$ in the standard G_2 root system, the $\lambda = +3$ cluster contains two short roots ($|\alpha| = \sqrt{2}$) and two long roots ($|\alpha| = \sqrt{6}$). The optimal grading assigns these different parities, achieving a lower value of $\hat{S}_g$ than any cluster-coherent partition (verified by exhaustive enumeration over all $2^6 = 64$ sign assignments on positive roots).

Remark (Scope of cluster coherence). The cluster coherence result (Theorem 6(i)) depends on the factorisation of interaction weights through eigenvalue clusters. This condition holds for all four classical Lie algebra families under standard Cartan decompositions (Corollary 8).

In exceptional algebras (G_2, F_4, E_6, E_7, E_8), eigenvalue clusters may contain roots of different lengths, leading to non-uniform interaction weights within a cluster. The functional then does not reduce to a bilinear form in cluster magnetisations, and optimal gradings may split clusters (Proposition 9). Extending the theorem to these cases requires refining the cluster structure to incorporate root-length data, yielding a multi-layer weighted max-cut problem. This generalisation is left for future work.

Corollary 10 (Higher-dimensional generalisation). *In $\mathfrak{sl}(5, \mathbb{R})$, a 4+1 Cartan element selects a (4, 1) grading. In $\mathfrak{sl}(6, \mathbb{R})$, a 5+1 element selects (5, 1), a 4+2 element selects (4, 2), and a 3+3 element selects (3, 3). All verified numerically across six spectral functionals. The mechanism is dimension-independent.*

Observation (Compact versus noncompact — empirical). In the defining representation with trace inner product, *compact* real forms ($\mathfrak{so}(n), \mathfrak{su}(n)$) yield negative $(\text{ad}_T^2)_{ii}$ on the active sector in all tested cases ($n = 3-7$), and the minimiser is the trivial grading $P = I$. The *split* (maximally noncompact) real forms $\mathfrak{sl}(n, \mathbb{R})$ yield positive $(\text{ad}_T^2)_{ii}$ on the active sector, and the minimiser is a nontrivial grading determined by the minority-cluster rule. For *non-split noncompact* forms ($\mathfrak{so}(p, q)$ with $p, q \geq 1$, $\mathfrak{sp}(2n, \mathbb{R})$), the sign depends on the Cartan direction: boost generators (noncompact) produce positive entries and nontrivial selection; rotation generators (compact) produce negative entries and trivial selection.

A general proof that compact $\Rightarrow$ negative active-sector eigenvalues would require specifying the inner product and representation in which $(\text{ad}_T^2)_{ii}$ is evaluated; we record the pattern as an empirical observation pending that analysis.

Type	Algebra	Compact?	ad_T^2 sign	Selected grading
A_3	$\mathfrak{sl}(4, \mathbb{R})$ with T_{B-L}	No	Positive	(3, 1)
A_4	$\mathfrak{sl}(5, \mathbb{R})$ with 4+1	No	Positive	(4, 1)
A_5	$\mathfrak{sl}(6, \mathbb{R})$ with 5+1	No	Positive	(5, 1)
B_2	$\mathfrak{so}(5)$	Yes	Negative	(5, 0) trivial
D_3	$\mathfrak{so}(6)$	Yes	Negative	(6, 0) trivial
D_2	$\mathfrak{so}(3, 1)$ boost	No	Positive	nontrivial
D_2	$\mathfrak{so}(3, 1)$ rotation	compact dir.	Negative	trivial
C_2	$\mathfrak{sp}(4, \mathbb{R})$	No	Positive	(2, 2)

Table 1: Grading selection across classical Lie algebra types. The sign of ad_T^2 refers to its diagonal entries on the active sector, computed in the defining (fundamental) representation with trace inner product $\langle X, Y \rangle = \text{Tr}(X^T Y)$. The sign pattern may depend on the choice of representation and inner product; all entries are empirical observations in the stated representation.

4 Numerical verification

The theorem is verified computationally in four independent tests. These jointly confirm that the minimiser is independent of the specific choice of spectral functional, coordinate frame, optimisation path, and small perturbations of the generator.

Test 1: Functional universality. Six spectral functionals ($f(x) = x^2, x^4, |x|, \mathbf{1}_{x \neq 0}, e^{x^2} - 1, \cosh x - 1$) all select (3, 1) as the unique minimum among the three grading classes, and as the global minimum on $\text{Gr}(1, 4)$ (5000-sample Monte Carlo confirmation, machine-precision agreement).

Test 2: Frame invariance. 50 random $\text{O}(4)$ conjugations $(T, P) \rightarrow (GTG^T, GPG^T)$ all preserve: (a) the (3, 1) grading selection (50/50), and (b) the alignment of P 's -1 eigenspace with T 's minority eigenvalue (50/50).

Test 3: Generator dependence and higher-dimensional generalisation. Table 2 reports the selected grading for Cartan elements across $\mathfrak{sl}(3, \mathbb{R})$ through $\mathfrak{sl}(6, \mathbb{R})$. In all 9 cases, the minimiser matches the weighted odd-support prediction of Theorem 6. Binary eigenvalue splits consistently select grading aligned with the minority cluster; ternary splits (e.g., 2+2+1 in $\mathfrak{sl}(5, \mathbb{R})$) are correctly handled by the weighted criterion of part (iii).

Test 4: Perturbation stability. Under perturbation $T_{B-L} \rightarrow T_{B-L} + \varepsilon H$ along five independent Cartan directions in $\mathfrak{sl}(4, \mathbb{R})$, the (3, 1) selection is structurally stable: the minimiser changes only at eigenvalue-crossing transitions ($\varepsilon \approx 0.3$ – 1.0 depending on direction) where the multiplicity structure transitions from 3+1 to 2+2. Perturbations that deepen the minority gap strengthen the selection monotonically with no bifurcation to $\varepsilon = 2$. Grading selection undergoes a discrete transition precisely at eigenvalue-crossing loci of the Cartan spectrum, where the multiplicity structure changes. This confirms the selection is not fine-tuned: any Cartan element in the 3+1 regime selects a (3, 1) grading.

Algebra	Cartan element	Eigenvalue structure	Selected grading
$\mathfrak{sl}(3, \mathbb{R})$	$(1/2, 1/2, -1)$	$2 + 1$	$(2, 1)$
$\mathfrak{sl}(4, \mathbb{R})$	$(1/3, 1/3, 1/3, -1)$	$3 + 1$	$(3, 1)$
$\mathfrak{sl}(4, \mathbb{R})$	$(1, 1, -1, -1)$	$2 + 2$	$(2, 2)$ split
$\mathfrak{sl}(5, \mathbb{R})$	$(1/4)^4, -1)$	$4 + 1$	$(4, 1)$
$\mathfrak{sl}(5, \mathbb{R})$	$(1/3)^3, (-1/2)^2)$	$3 + 2$	$(3, 2)$
$\mathfrak{sl}(5, \mathbb{R})$	$(1, 1, -1, -1, 0)$	$2 + 2 + 1$ (ternary)	$(3, 2)$
$\mathfrak{sl}(6, \mathbb{R})$	$(1/5)^5, -1)$	$5 + 1$	$(5, 1)$
$\mathfrak{sl}(6, \mathbb{R})$	$(1/4)^4, (-1/2)^2)$	$4 + 2$	$(4, 2)$
$\mathfrak{sl}(6, \mathbb{R})$	$(1/3)^3, (-1/3)^3)$	$3 + 3$	$(3, 3)$

Table 2: Grading selected by $\tilde{\mathcal{S}}_g$ with $g(x) = x^2$ for Cartan elements across $\mathfrak{sl}(3, \mathbb{R})$ – $\mathfrak{sl}(6, \mathbb{R})$. All 9 cases match the weighted odd-support prediction of Theorem 6. The 2+2+1 case in $\mathfrak{sl}(5, \mathbb{R})$ is a ternary split correctly handled by the weighted criterion: the strongest inter-cluster generators (gap = 2) dominate over weaker ones (gap = 1).

5 Interpretation and scope

The mathematical result. Theorem 6 is a statement about $\mathbb{Z}/2$ gradings of a Lie algebra: given a Cartan element with a binary eigenvalue split, the grading that makes all cross-cluster root spaces odd is uniquely selected by minimisation of a universal class of spectral functionals. For T_{B-L} in $\mathfrak{sl}(4, \mathbb{R})$, this grading has the $(3, 1)$ shape. The result is combinatorial and exact.

What the grading acts on. The involution P and the grading $\mathfrak{g}_+ \oplus \mathfrak{g}_-$ it induces act on the *internal carrier space* $\mathbb{R}^4$ of the algebra $\mathfrak{sl}(4, \mathbb{R})$, which in the Pati–Salam context is the $B-L$ charge space (three quark colours plus one lepton). The 3+1 split selected by the theorem is a 3+1 decomposition of this *internal* space. It is *not*, by itself, a derivation of the $(3, 1)$ Lorentzian signature of physical spacetime.

The Coleman–Mandula constraint. The Coleman–Mandula theorem [3] and its Haag–Łopuszański–Sohnius refinement [4] constrain the mixing of internal and spacetime symmetries in any relativistic quantum field theory with a mass gap: the full symmetry group must factorise as (internal) $\times$ (Poincaré), up to possible supersymmetric extensions. Identifying the internal 3+1 charge-sector grading selected by Theorem 6 with the 3+1 Lorentzian signature of spacetime would constitute precisely such a mixing, and is therefore not permissible without additional structure.

What would be needed to bridge the gap. Three routes could, in principle, connect the internal grading to spacetime signature without violating Coleman–Mandula:

1. *Soldering construction.* In the Palatini formulation, the tetrad e_μ^a “solders” the internal $\mathfrak{sl}(4, \mathbb{R})$ indices to spacetime indices. If the soldering form is present, the internal grading induces a spacetime grading. But this requires the soldering form as additional input — it is not produced by the algebraic minimisation alone.
2. *Pre-geometric regime.* If the construction operates at a level where Poincaré symmetry has not yet emerged (e.g., a pre-geometric or background-independent formulation), the Coleman–Mandula hypotheses do not apply and the internal/spacetime distinction is not yet defined. The grading could then be interpreted as a *proto-signature* that seeds the eventual spacetime structure upon emergence of a geometric phase.

3. *Explicit evasion.* If the theory is formulated with massless particles only, or in a way that violates other Coleman–Mandula hypotheses (e.g., infinite-dimensional symmetry, absence of a mass gap), the no-go theorem does not apply and the internal-spacetime identification could be direct.

None of these routes is established in the present work. The algebraic result (Theorem 6) stands on its own as a grading-selection theorem; the physical interpretation as spacetime Adjoint Spectral Minimization and Bipartite Adjoint Spectral Minimization and Bipartite Adjoint Spectral Minimization and Bipartite Adjoint Spectral Minimization and Bipartite Signature Selection in Clifford Algebras in Clifford Algebras in Clifford Algebras in Clifford Algebras remains an open conjecture requiring one of the above bridging mechanisms.

The honest scope of the result. The theorem selects a 3+1 grading of the internal carrier space of $\mathfrak{sl}(4, \mathbb{R})$ associated with the $B-L$ generator. This grading has the same shape as the Lorentzian signature of spacetime. Whether this coincidence reflects a deep structural connection (via one of the bridging mechanisms above) or is merely a numerical coincidence of the $n = 4$ case is not determined by the algebraic argument alone.

In coordinate-free language, the mathematical content is:

The minimiser is the grading that makes all active cross-cluster root spaces odd. For T_{B-L} , this grading has $(3, 1)$ shape.

Compact versus noncompact algebras. The mechanism extends beyond the A_n series. Numerical tests on compact orthogonal algebras $\mathfrak{so}(n)$ ($n = 3, \dots, 6$), the noncompact Lorentz algebra $\mathfrak{so}(3, 1)$, and the symplectic algebra $\mathfrak{sp}(4, \mathbb{R})$ confirm the general pattern (16 cases total, all consistent):

- *Compact* algebras have $\text{ad}_T^2 < 0$ on the active sector. The minimiser assigns all active generators to the even subspace.
- *Noncompact* algebras have $\text{ad}_T^2 > 0$ on the active sector (or mixed signs in algebras like $\mathfrak{so}(3, 1)$). The minimiser assigns positive- ad_T^2 generators to the odd subspace, producing a nontrivial grading.

This parallels the standard relationship between real forms and the Cartan decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$; whether the two decompositions coincide in general is an open question.

What is not claimed.

1. The result selects a grading of an *internal* carrier space, not of physical spacetime. Bridging the two requires additional structure (soldering, pre-geometry, or Coleman–Mandula evasion) not provided here.
2. The result does not explain SM parity violation ($SU(2)_L$ left-handedness). That requires the full Pati–Salam product group $SU(4)_C \times SU(2)_L \times SU(2)_R$, not $\mathfrak{sl}(4, \mathbb{R})$ alone.
3. The minimisation principle $\delta \tilde{\mathcal{S}}_g = 0$ is proposed, not derived from a deeper action. It is the simplest consistent choice, but “simplest” is a criterion, not a proof.
4. The mechanism has been verified across all four classical types (A_n, B_n, C_n, D_n) and is dimension-independent. Extension to exceptional algebras ($G_2, F_4, E_{6,7,8}$) remains open.

Remark (Non-commutativity interpretation). The selection mechanism is equivalent to assigning negative grading to the minimal index set that maximises the measure of non-commuting directions under ad_T .

Acknowledgment

This result emerged from an extended investigation of chirality structure in the ACS bracket algebra. An initial measurement of a “72% parity-odd holonomy bias” was classified as grading-induced after invariance testing. The upgrade to a grading-selection theorem occurred when the spectral functional $\widehat{\mathcal{S}}_g = \text{Tr}(\sigma_P \cdot g(\text{ad}_T))$ was found to select a unique $(3, 1)$ grading. Subsequent universality testing (six functionals, frame invariance, generator dependence, higher-dimensional generalisation) confirmed the robustness of the selection. A physicist’s review identified the need to restrict to $g(0) = 0$ for a clean theorem statement, to separate the $f(0) > 0$ case as a distinct proposition, and — most importantly — to distinguish the internal grading result from any claim about physical spacetime signature, in light of the Coleman–Mandula theorem. These corrections significantly strengthened the paper.

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